FleetCode: Autonomous Sailboats for Cargo Delivery

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Objective: Design a fleet of efficient, small **autonomous** sailboats, with **object detection** and **collision avoidance**, **dynamic repathing**, swarm behavior, and inexpensive modular design

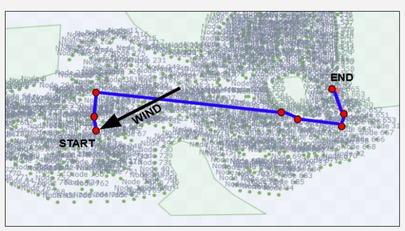
Motivation

- Large cargo ships are expensive, bulky, and easily targeted
- >1000 casualties at sea each year on shipping vessels
- Net-zero energy solutions are ideal for low priority cargo
- **No existing solutions for autonomous ocean cargo transport**

Design Criteria

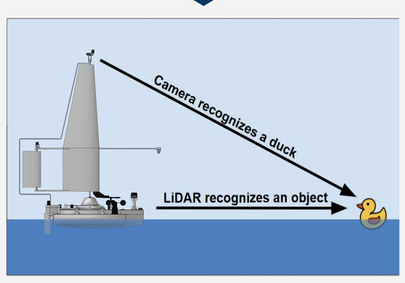
Fleet Design	Coordinated dual-vessel operation	✅
Smaller Size	15 lb payload capacity	✅
Net-Zero Energy	Wind-driven, solar-powered	✅
Autonomous	Self-guided navigation	✅
Object Avoidance	Dynamic repathing around obstacles	✅
Modularity	2 person set up in < 10 min	✅
Low Cost	Total boat cost of < $1,000	✅

Autonomous Sailing



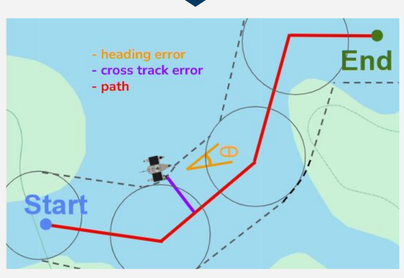
Global Path Planning

- Computes best path with modified A-Star algorithm.
- Route adjusts for wind conditions with optimized directional changes.



Live Repathing

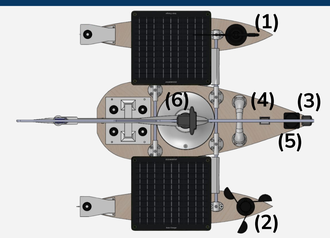
- Camera + Light Detection and Ranging (LiDAR) detect obstacles and update maps.
- Repathing creates updated path around newly found obstacles.



Low-Level Control

- Navigation algorithm calculates cross-track and heading errors.
- Proportional Integral (PI) controller adjusts rudder to follow path near waypoints.

Electrical and Mechanical Design



Sensor Suite

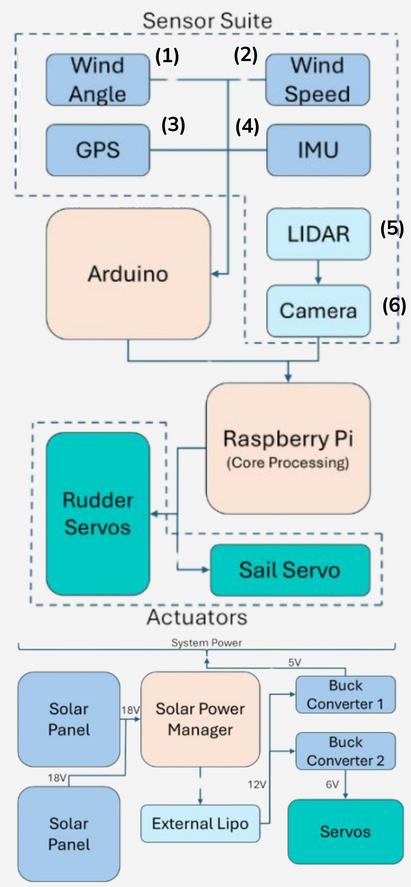
- Wind direction, wind speed (anemometer), GPS, LiDAR, and IMU for real-time measurements
- Sensor fusion algorithm for precise state estimation and dynamic response

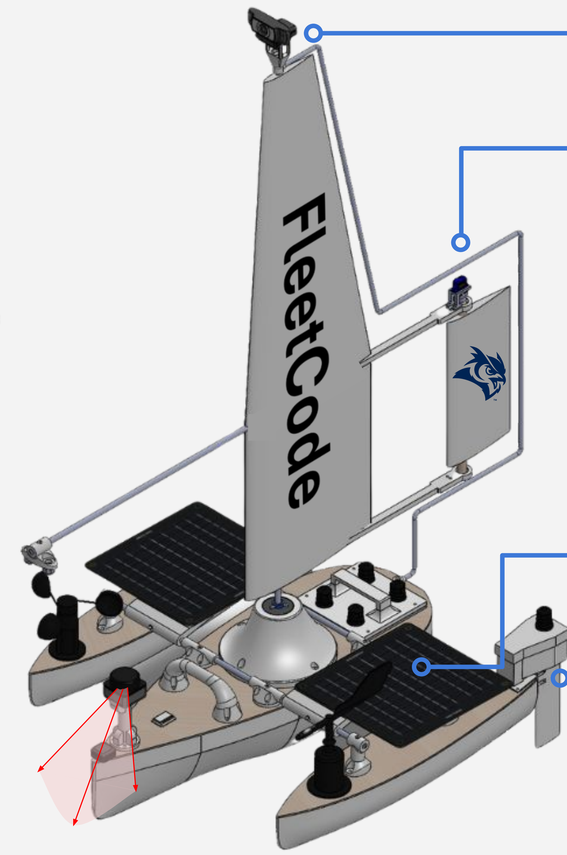
Trimaran Design

- Outriggers provide strong righting moment to ensure stability
- Hulls run thin in the water to balance straight line drag and prevent drift without needing a keel

Construction Methods

- Using low cost materials
  - Plywood (hull lids)
  - 3D Printed PLA (outriggers and main hull)
  - Fiberglass and resin, plus marine sealant for strength and waterproofing





CAD diagram of Fleetcode boat.

Computer Vision

- Integrated object identification with distance estimation and global heading for path updates

Optimal Sail Design

- Small elevator adjustment allow for sail actuation to generate force
- Lift-drag distribution was validated using computational fluid dynamic simulation to find optimal sail angles

Modular Design

- Design uses non permanent fasteners for ease of disassembly
- Parts are easily 3D printed with minimal support for easy parts replacement

Solar Panels

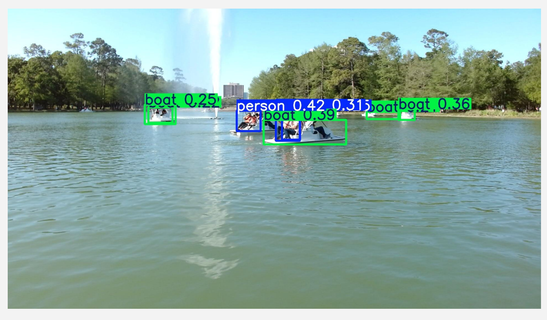
- Provides supplemental power during sunny conditions to optimize low energy consumption

Hydrodynamic Rudder

- 3D printed and fiberglassed hydrofoils
- 2x1 spur gearbox to reduce backdriving
- Two rudders for redundancy and enhanced maneuverability

Testing Results

- ***Global Pathing***: Quickly determined fastest accurate path to destination.
- ***Precise Control***: Precisely navigated multiple consecutive waypoints.
- ***Computer Vision***: Detected objects and their position relative to the boat.
- ***Hydrodynamic Performance***: Stable in various wind and wave conditions.



Computer vision detecting obstacles.

Path during test at McGovern lake.

Conclusion

- Fleetcode offers an **efficient, inexpensive alternative** to manned vessels for military cargo operations.
  - Reduces carbon footprint with **energy-efficient** vessels
  - **Decentralized design** adapts swiftly to different needs
  - **Autonomous operation** eliminates need for human crews
- **Next Steps:**
  - Improve obstacle avoidance, optimize repathing algorithm
  - Boost fleet coordination, expand fleet with additional units

References and Acknowledgments

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